

Digital Signal Analysis and Processing Exam Notes

CT 704 (BEI/BCT, IV-I) / EX 753 (BEX, IV-II) · TU, IOE · tiers over 45 papers, 2066–2082 BS

7. Discrete Fourier Transform and FFT

7 hrs · 99 questions across the 45 papers · worth about 14 of 80 marks · 645 marks, 17.6% of the archive · 593 numerical (92%) + 52 theory · 3 syllabus sub-topics · the heaviest chapter in the subject.

Read this first:

- **Three questions, and only three**, in 44 of the 45 papers:
 - **Compute an N -point DFT by DIT or DIF** TOP 43/45, 340 marks. Almost always **8-point**, almost always "draw the butterfly".
 - **Circular convolution** TOP 39/45, 253 marks, in three disguises that are all the same calculation (\$7.5).
 - **Bookwork on the DFT** TOP 17/45, 52 marks: why DFT, why FFT, the properties, zero padding.
- **The FFT is not a new transform**. It is the DFT, computed by a rearrangement that costs $\frac{N}{2} \log_2 N$ multiplications instead of N^2 . Anyone who says "the FFT of $x[n]$ " means its DFT.
- **DIT and DIF give identical answers**. They differ only in where the bit-reversal sits and where the twiddle factor is applied. If a paper does not say which, either is a full-marks answer.
- **Everything rests on two identities**, and nothing else:

$$W_N = e^{-j2\pi/N}, \quad W_N^{k+N/2} = -W_N^k, \quad W_N^{k+N} = W_N^k$$

The first halves the work at every stage; the second is what makes the recursion close.

- **A sequence shorter than N is zero padded**. Eleven papers print six or seven values for an 8-point transform. That is not a misprint to fix, it is the zero-padding they ask you to define elsewhere in the same paper.

7.1 What the DFT is, and why it exists

Why do we need the DFT / FFT although we have the DTFT? Differentiate between the DFT and the DTFT 1+5 2+4 6 2

TOP 13/45 79 Bh, 78 Bh, 71 Ch, 81 Bh, 79 Ch, 70 Bh, 79 Ba, 80 Ba, 74 Ma · asked in nine papers, always the same two reasons

Why the DTFT is not enough

- The DTFT $X(e^{j\omega}) = \sum_n x[n]e^{-j\omega n}$ is a **continuous** function of ω .
- **A computer cannot store a continuous function**, and it cannot sum an infinite series. Neither end of the DTFT is computable.
- The DFT fixes both at once: **finite** sequence in, **finite** set of samples out.

$$X[k] = \sum_{n=0}^{N-1} x[n]W_N^{nk}, \quad x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k]W_N^{-nk}$$

- **The DFT is the DTFT sampled at N equally spaced points**, $\omega_k = 2\pi k/N$. Nothing more.
- The price: the DFT implicitly treats $x[n]$ as **one period of a periodic sequence**. Every oddity in this chapter, circular shift, circular convolution, wrap-around, follows from that one fact.

DFT against DTFT, as a table

	DTFT	DFT
Input	any length, even infinite	finite , N points
Output	continuous in ω	discrete , N points
Periodic in	ω , period 2π	k , period N
Computable	no	yes
Assumes	nothing	$x[n]$ is periodic with period N
Convolution	linear	circular

And why the FFT

- The DFT as defined costs N^2 complex multiplications. **The FFT is an algorithm for the same numbers** at $\frac{N}{2} \log_2 N$.
- For $N = 8$: 64 against 12. For $N = 1024$: 1 048 576 against 5120, a **speed-up of 204.8**. Quote a number; it is worth a mark.

How does the FFT reduce computational complexity compared with direct computation of the DFT?

5 2 4+2

TOP 12/45 the same nine papers, plus 74 Bh 2, 73 Ma 2, 79 Ch 4+2

The argument, in four lines

1. Direct DFT: N outputs, each a sum of N products. N^2 **complex multiplications and $N(N-1)$ additions.**
2. Split $x[n]$ into two half-length sequences. Two $\frac{N}{2}$ -point DFTs cost $2(\frac{N}{2})^2 = \frac{N^2}{2}$, **half the work**, plus $\frac{N}{2}$ twiddle multiplications to recombine.
3. **Repeat the split.** $N = 2^m$ allows it $m = \log_2 N$ times, until only 2-point DFTs remain, and a 2-point DFT needs **no multiplication at all** ($W_2^0 = 1$).
4. Each of the $\log_2 N$ stages has $\frac{N}{2}$ butterflies and each butterfly costs one multiplication:

$$\frac{N}{2} \log_2 N \text{ multiplications, } N \log_2 N \text{ additions}$$

The numbers to quote

N	Direct N^2	FFT $\frac{N}{2} \log_2 N$	Speed-up
8	64	12	5.33
32	1024	80	12.8
256	65 536	1024	64
1024	1 048 576	5120	204.8

- **The saving grows with N** , which is the point: it is $2N/\log_2 N$, not a constant factor.
- **Radix-2 needs N a power of 2.** If it is not, zero pad up to one. That is the second reason zero padding exists.
- The two properties being exploited are **periodicity** $W_N^{k+N} = W_N^k$ and **symmetry** $W_N^{k+N/2} = -W_N^k$. Name them: they are usually worth a mark each.

7.2 Properties of the DFT

State and prove the circular convolution property of the DFT; write about the multiplication and convolution properties 2+2+4 6 1

HOT 4/45 70 Ch 2+2+4, 69 Bh 6, 75 Ash 1, 73 Bh 1 · the proof is five lines

The property, and its proof

- **Statement.** If $x_3[n] = x_1[n] \otimes x_2[n]$, the *circular convolution*, then

$$X_3[k] = X_1[k] X_2[k]$$

Circular convolution in time is multiplication in frequency.

- **Proof.** Write the circular convolution and transform it:

$$X_3[k] = \sum_{n=0}^{N-1} \left(\sum_{m=0}^{N-1} x_1[m] x_2[((n-m))_N] \right) W_N^{nk}$$

- Swap the sums and put $\ell = n - m$, so $n = \ell + m$ and $W_N^{nk} = W_N^{mk} W_N^{\ell k}$:

$$= \sum_{m=0}^{N-1} x_1[m] W_N^{mk} \sum_{\ell=0}^{N-1} x_2[((\ell))_N] W_N^{\ell k} = X_1[k] X_2[k]$$

- **The modulo- N index is what makes the inner sum come out as $X_2[k]$ exactly**, with no leftover terms. That is the whole proof, and the step worth pointing at.
- **The dual (multiplication property):** multiplying in time is *circular convolution divided by N* in frequency, $x_1[n] x_2[n] \longleftrightarrow \frac{1}{N} X_1[k] \otimes X_2[k]$. **Do not forget the $1/N$.**

The full property table (Oppenheim Table 8.2)

Sequence, length N	N -point DFT
$x[n]$	$X[k]$
$ax_1[n] + bx_2[n]$	$aX_1[k] + bX_2[k]$
$X[n]$	$Nx[(-(k))_N]$
$x[((n-m))_N]$	$W_N^{km} X[k]$
$W_N^{-\ell n} x[n]$	$X[((k-\ell))_N]$
$\sum_m x_1[m] x_2[((n-m))_N]$	$X_1[k] X_2[k]$
$x_1[n] x_2[n]$	$\frac{1}{N} \sum_{\ell} X_1[\ell] X_2[((k-\ell))_N]$
$x^*[n]$	$X^*[((-k))_N]$
$x[n]$ real	$X[k] = X^*[((-k))_N]$

- **Row 4 is the circular shift:** a shift in time is a *phase* in frequency and nothing else. The magnitude spectrum does not move.
- **Row 3 is duality**, and it is the whole of the 67 Mng question in §7.3.
- The last row is the free check on every answer in this chapter: **a real sequence has a conjugate-symmetric spectrum**, so $X[N-k] = X^*[k]$. Compute half the values and mirror the rest.

Define zero padding, and Parseval's theorem 1 2

TOP 7/45 74 Ch, 74 Ash, 70 Ch, 81 Ch, 75 Bh, 74 Bh 1; 77 Ch 2 · eight free marks

Zero padding

- **Appending zeros** to a sequence before taking its DFT, to reach a chosen length N .
- **Three reasons**, and papers accept any of them:
 - to reach a **power of 2** so a radix-2 FFT can run;
 - to make $N \geq L + M - 1$ so a **circular convolution equals the linear one** (§7.5, and this is the reason the question sits beside those problems);
 - to **interpolate the spectrum**: more samples of the same underlying DTFT.
- **Zero padding does NOT add information and does NOT improve resolution**. It samples the same DTFT more finely. Saying otherwise loses the mark.

Parseval's theorem

- Energy is the same computed either side of the transform:

$$\sum_{n=0}^{N-1} |x[n]|^2 = \frac{1}{N} \sum_{k=0}^{N-1} |X[k]|^2$$

- **Note the $1/N$** , which is on the frequency side, matching the IDFT.
- Useful as an arithmetic check: compute $\sum |x[n]|^2$ from the given sequence in two seconds, then confirm the $X[k]$ you just worked out.

Linear against circular convolution (75 Ch asks for it directly)

- **Linear**: indices run freely, result has length $L + M - 1$, and it is what a real LTI system does.
- **Circular**: indices are taken mod N , result has length N , and it is what multiplying two DFTs gives you.
- **They agree exactly when $N \geq L + M - 1$** , and otherwise the circular one is the linear one **wrapped round and added to itself**.

7.3 The IDFT from a forward DFT

Using the duality property, how can a forward DFT routine be used to compute the IDFT? 8 2+6

PIN 2/45 67 Mng 8, 72 Ash 2+6 · 8 marks for four lines of algebra

The conjugate trick, which is the expected answer

1. Start from the IDFT and conjugate both sides:

$$x[n] = \frac{1}{N} \sum_k X[k] W_N^{-nk} \Rightarrow x^*[n] = \frac{1}{N} \sum_k X^*[k] W_N^{nk}$$

2. **The right-hand sum is a forward DFT of $X^*[k]$** . So:

$$x[n] = \frac{1}{N} \left(\text{DFT} \{ X^*[k] \} \right)^*$$

3. **Three steps**: conjugate the input, run the same FFT, conjugate the output and divide by N .
4. **No second routine is needed**, which is the point of the question: the same butterfly hardware computes both directions.

The duality route, if the paper insists

- Duality says $X[n] \xrightarrow{\text{DFT}} Nx[((-k))_N]$.
- So feeding the *spectrum* $X[k]$ into a forward DFT returns the original sequence, scaled by N and **index reversed modulo N** :

$$x[n] = \frac{1}{N} \text{DFT} \{ X[k] \} \Big|_{k \rightarrow -n \bmod N}$$

- Index reversal modulo N leaves entry 0 alone and reverses the rest, so $\{a, b, c, d\} \rightarrow \{a, d, c, b\}$. **That single sentence is what the 8 marks are for.**
- Both routes are correct. The conjugate one is easier to defend and is what 72 Ash's worked answer needs (§2 of the numericals).

7.4 The FFT: decimation in time and in frequency

How do you use the DIT-FFT algorithm to compute the 8-point DFT of a sequence? Explain with the flow graph **6 4**

TOP 43/45 70 Ma **6**, 69 Bh **4** (4-point), and the 41 computation papers in \$7.6

Decimation in time, derived

- Split the sum by **even and odd n** :

$$X[k] = \sum_r x[2r]W_N^{2rk} + \sum_r x[2r+1]W_N^{(2r+1)k}$$

- $W_N^2 = W_{N/2}$, so both sums are $\frac{N}{2}$ -point DFTs:

$$X[k] = G[k] + W_N^k H[k]$$

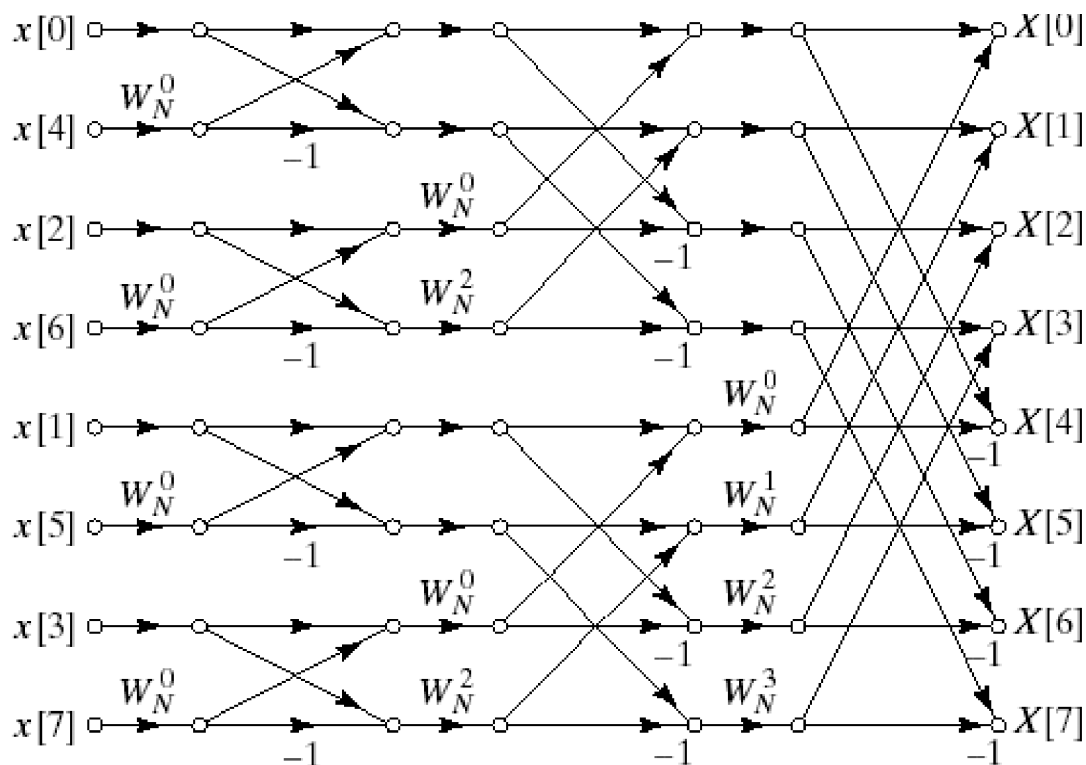
with G from the even samples and H from the odd ones.

- The other half comes free from the symmetry:

$$X[k + \frac{N}{2}] = G[k] - W_N^k H[k]$$

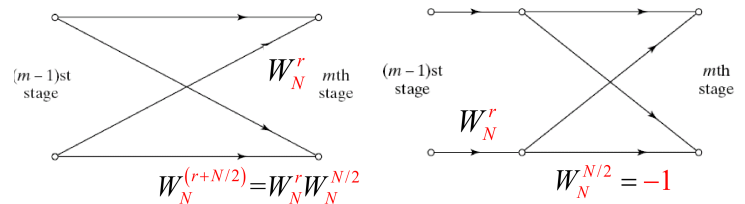
- One multiplication serves two outputs.** That pair is the butterfly, and it is the entire saving.
- Recurse $\log_2 N$ times. Splitting by even and odd repeatedly leaves the input in **bit-reversed order**; the output comes out natural.

The 8-point DIT flow graph (**draw this one**)



Input bit-reversed $\{x[0], x[4], x[2], x[6], x[1], x[5], x[3], x[7]\}$, output natural. Three stages. Twiddles W_N^0 in stage 1; W_N^0, W_N^2 in stage 2; $W_N^0, W_N^1, W_N^2, W_N^3$ in stage 3. The twiddle count doubles each stage and the butterfly span doubles with it.

The butterfly



Left, the butterfly as first drawn: two twiddles. Right, the same butterfly after $W_N^{r+N/2} = W_N^r W_N^{N/2} = -W_N^r$ is used, so one multiplication and a sign.

Always draw the right-hand form.

- Every stage is $\frac{N}{2}$ of these; there are $\log_2 N$ stages.
- The -1 on the lower branch is not decoration.** Omitting it is the single most common way to lose the marks on a flow graph.

Draw the flow graph for the 8-point decimation in frequency algorithm 8 2+6

TOP 20/45 the DIF half of §7.6: 80 Bh, 76 Ch, 76 Ash, 79 Ba, 73 Ch, 69 Ch, 74 Ash, 70 Bh, 81 Ch, 78 Ch, 77 Ch, 75 Bh, 72 Ma, **68 Bh**, 82 Ba, 80 Ba, 70 Ch, 74 Ch

Decimation in frequency, derived

- This time split the **input** into first and second halves, and the **output** into even and odd k :

$$g[n] = x[n] + x[n + \frac{N}{2}], \quad h[n] = (x[n] - x[n + \frac{N}{2}])W_N^n$$

$$X[2r] = \text{DFT}_{N/2}\{g[n]\}, \quad X[2r+1] = \text{DFT}_{N/2}\{h[n]\}$$

- **The twiddle multiplies the DIFFERENCE, after the subtraction.** In DIT it multiplies before the add. That is the only real difference between the two.
- Recurse $\log_2 N$ times. Input stays natural; **output comes out bit-reversed** and has to be unscrambled.

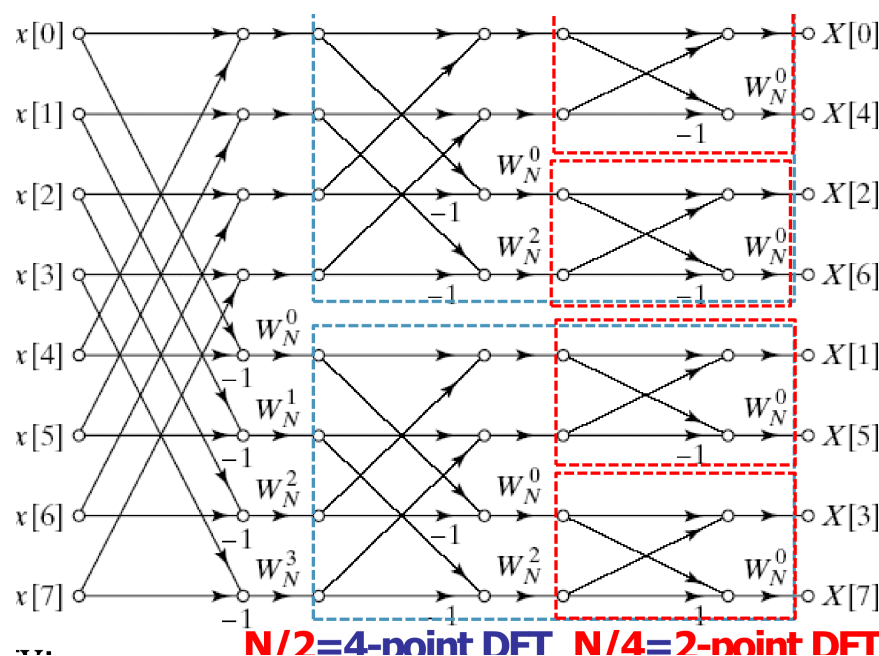
Bit reversal, which is all either algorithm scrambles

- **The same permutation serves both:** 0, 4, 2, 6, 1, 5, 3, 7 for $N = 8$. It is its own inverse, so there is only one order to memorise.
- Get it by writing the index in binary and reading it **backwards**: $3 = 011 \rightarrow 110 = 6$, $1 = 001 \rightarrow 100 = 4$.
- **Exactly one end is scrambled**, never both and never neither. DIT scrambles the input, DIF the output.
- For $N = 4$ it is 0, 2, 1, 3; for $N = 16$, 0, 8, 4, 12, 2, 10, 6, 14, 1, 9, 5, 13, 3, 11, 7, 15.
- The scrambling is not overhead you can avoid. It is what lets every butterfly write its result back over its own inputs, so the whole transform runs in **one array with no scratch space**.

DIT against DIF, the whole comparison

	DIT	DIF
Splits	input, even/odd n	output, even/odd k
Input order	bit-reversed	natural
Output order	natural	bit-reversed
Twiddle	before the add	after the subtract
Stages	$\log_2 N$	$\log_2 N$
Multiplies	$\frac{N}{2} \log_2 N$	$\frac{N}{2} \log_2 N$
Answer	identical	

The 8-point DIF flow graph (the other one to be able to draw)



Input natural, **output bit-reversed**: $X[0], X[4], X[2], X[6], X[1], X[5], X[3], X[7]$ read down the right edge. The dashed boxes show the recursion, the $N/2 = 4$ -point DFTs inside the whole transform and the $N/4 = 2$ -point DFTs inside those. The twiddles run W_N^0 to W_N^3 in stage 1 and halve their range at each stage, the mirror image of DIT, where they start at W_N^0 alone and double.

7.5 Circular convolution, and the three ways it is asked

Compute the circular convolution of two sequences 5 6 7 2+6

TOP 39/45 15 papers ask it directly, 12 ask for $x_3 = \text{IDFT}(X_1 X_2)$, 4 ask for a linear convolution done circularly. All 31 are one calculation.

In one line: **circular convolution is ordinary convolution with the index taken modulo N** , so the tail that would have run off the end wraps round and adds onto the front instead. Everything below is that sentence applied.

The three disguises

1. "Compute the circular convolution of x_1 and x_2 ." Direct.
2. "Find $x_3[n]$ if $X_3[k] = X_1[k]X_2[k]$." By the property in §7.2 this **IS the circular convolution**, so do the same arithmetic and do not compute a single DFT.
3. "Find the linear convolution using circular convolution." Zero pad both to $N \geq L + M - 1$, then the same arithmetic again.

Recognising that (2) needs no transform is worth several minutes in the exam. The long route, three DFTs and an IDFT, earns the same marks and takes four times as long. Say in one line that you are using the property, then convolve.

Choosing N

- If the paper names N , use it, and pad both sequences up to it.
- If not, $N = \max(L, M)$ for a circular answer, and $N = L + M - 1$ for a linear one.
- **N smaller than $L + M - 1$ wraps**, and the wrap is not an error: it is the correct circular answer to the question that was asked.
- **N larger than $L + M - 1$ only appends zeros** to the result. Harmless, and sometimes needed to reach a power of 2.

The matrix method, which is the fastest by hand

- Write the **circulant** of one sequence: first column is x_2 , and each next column is the previous rotated *down* by one.

$$y[n] = \sum_{m=0}^{N-1} x_1[m] x_2[(n-m)_N]$$

- Multiply by x_1 as a column vector. **Each row of the product is one output sample**, so the whole answer falls out of one $N \times N$ table.
- Check: $\sum y[n] = (\sum x_1)(\sum x_2)$, always. It is $Y[0] = X_1[0]X_2[0]$ in disguise, it costs two additions, and it catches almost every slip.

The concentric-circle picture

- The same arithmetic, drawn: x_1 clockwise on an inner circle, x_2 anticlockwise on an outer one, multiply the aligned pairs and sum, then rotate one circle a step for the next output.
- **Use the matrix instead.** The circles are slower, easier to misalign, and no paper in the 45 demands them.

7.6 Last-minute recall

Say these without thinking

- $X[k] = \sum_{n=0}^{N-1} x[n] W_N^{nk}$, $W_N = e^{-j2\pi/N}$, IDFT carries $1/N$ and W_N^{-nk} .
- W_4 : 1, $-j$, -1 , $+j$. $W_8^1 = \frac{1-j}{\sqrt{2}}$, $W_8^2 = -j$, $W_8^3 = \frac{-1-j}{\sqrt{2}}$.
- $W_N^{k+N/2} = -W_N^k$ and $W_N^{k+N} = W_N^k$.
- Bit reversal for $N = 8$: 0, 4, 2, 6, 1, 5, 3, 7.
- FFT cost $\frac{N}{2} \log_2 N$ multiplies, $N \log_2 N$ adds; $N = 1024$ gives $204.8 \times$.
- DIT: bit-reversed in, natural out, twiddle before the add. DIF: natural in, bit-reversed out, twiddle after the subtract.
- $X_3[k] = X_1[k]X_2[k] \Leftrightarrow$ **circular** convolution.
- Linear = circular when $N \geq L + M - 1$.
- $X[0] = \sum x[n]$, and for real x , $X[N-k] = X^*[k]$.
- Parseval: $\sum |x[n]|^2 = \frac{1}{N} \sum |X[k]|^2$.

Mistakes that cost marks

- **Forgetting the -1** on the lower butterfly branch.
- Feeding DIT in natural order, or reading DIF's output in natural order. **Exactly one of the two ends is scrambled, never both and never neither.**
- Using $e^{+j2\pi/N}$ in the forward transform. **The forward sign is minus.**
- Computing three DFTs and an IDFT for a question that only wanted a circular convolution.
- Treating a 6 or 7 entry sequence as an error. **Pad it and say you padded it.**
- Claiming zero padding improves frequency resolution. It does not.
- Dropping the $1/N$ in the IDFT, or in the multiplication property.
- Giving a circular convolution of length $L + M - 1$. **It has length N .**
- Forgetting that $X[k]$ repeats: $X[5]$ of a 4-point DFT is $X[1]$, which is exactly what 75 Ch is testing.

Abbreviations used in these notes

Only these are ever shortened. Anything not on this list is written out in full. → reads “leads to / therefore”, + reads “and”, = reads “is”. ✓ means the property holds, × that it does not.

Short	Means	Short	Means
DSP	digital signal processing	DSAP	digital signal analysis and processing
DT	discrete-time	CT	continuous-time
LTI	linear time-invariant	LSI	linear shift-invariant, the same thing
TI	time-invariant	BIBO	bounded input, bounded output
LCCDE	linear constant-coefficient difference equation	ROC	region of convergence
DTFS	discrete-time Fourier series	DTFT	discrete-time Fourier transform
CTFT	continuous-time Fourier transform	DFS	discrete Fourier series
DFT	discrete Fourier transform	IDFT	inverse DFT
FFT	fast Fourier transform	DIT	decimation in time
DIF	decimation in frequency	MAC	multiply and accumulate
FIR	finite impulse response	IIR	infinite impulse response
LPF	low pass filter	HPF	high pass filter
BPF	band pass filter	BSF	band stop filter
BLT	bilinear transformation	II	impulse invariance
A/D	analog to digital	D/A	digital to analog
S/H	sample and hold	ZOH	zero-order hold
SNR	signal to noise ratio	PYQ	past-year question
coeff	coefficient	freq	frequency
eqn	equation	no.	number
ops	operations	attr	attribute
w/	with	w/o	without
Eg	for example	Vs	compare / differentiate
+Fig	a figure is wanted	(I)	importance / need
(+)	advantage	(-)	disadvantage

Exam paper codes

Month codes

- **Ba** Baishakh **Jth** Jestha **Asa** Ashad
- **Shr** Shrawan **Bh** Bhadra **Ash** Ashwin
- **Ka** Kartik **Mng** Mangsir **Po** Poush
- **Ma** Magh **Ch** Chaitra
- **Ash and Asa are different months**: 2075 Ashwin and 2070 Ashad are separate papers.

How a year code is written

- **Bold** = **Regular** exam. Plain = Back exam.
 - “New Back (2066 & Later Batch)” counts as Back.
 - A header printing “Regular / Back” in one box counts as Regular.
- Typewriter = the EX 753 paper (DSP, BEX, IV/II). Normal font = **CT 704** (DSAP, BEI/BCT, IV/I).
- The two axes are independent: **76 Ch** is CT 704 Regular, 76 Ash is CT 704 Back, 76 **Bh** is EX 753 Regular, 74 **Ma** is EX 753 Back.
- **69 Bh and 69 Bh** are two different papers, one in each course.

Frequency chips

- **TOP n/45** asked in **10 or more** of the 45 papers.
- **HOT n/45** asked in **4 to 9** papers.
- **PIN n/45** asked in **1 to 3** papers.
- **verified/added** written or corrected against a reference because the source notes did not cover it, or covered it wrongly.
- **5** marks the question has actually carried. Several chips mean the same question has been set at different marks in different years.
- The count is over **45 papers**: 28 CT 704 (2066 Magh to 2082 Bhadra) and 17 EX 753 (2069 Bhadra to 2081 Chaitra).
- Counts come from the Detailed PYQ document, which reproduces every question in full.